

98 Per Cent of the Procedure, 26 Per Cent of the Code: The Enacted Law and the Applied Law in 50,666 Judgments

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Abstract

Saudi commercial courts have cited 97.9 per cent of the articles of the law telling them how to proceed, and 25.8 per cent of the code telling them what the parties owe each other. The contrast is drawn from 106,337 statutory citations in 50,666 published judgments, matched to 15,855 articles across 291 instruments. 11.2 per cent of the statute book is ever cited, 29.4 per cent within the courts' jurisdiction; what is applied is 88.7 per cent procedural. Segmenting judgments by their headings locates the narrowing at the bench, not the bar. Amendment predicts neither citation nor its absence.

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JEL codes: K40, K41, K10, C81

1 Introduction

A legislature enacts. A court applies. The two facts are studied by different literatures, using different data, and are rarely put side by side — not because the question is uninteresting but because answering it requires two corpora for the same jurisdiction, at the same level of granularity, and those seldom exist together. Legislation is studied where consolidated statutory text is machine-readable, chiefly the United States and Germany. Judicial behaviour is studied where opinions are digitised at scale, chiefly the same places. Where both exist, they are usually studied apart.

This article joins them for Saudi Arabia, a jurisdiction that has recently codified much of its private and commercial law and that publishes its commercial judgments in full text. On one side is a verified corpus of 15,855 articles across 291 instruments, each article carried with its number, its text and its legal status. On the other are 50,666 judgments, published by the Ministry of Justice, from which 106,337 statutory citations are extracted and matched to the article they cite. The result is a link between the enacted and the applied at the level at which a lawyer actually argues: not the statute, the article.

What that link shows is a legal system using a small and unrepresentative part of what it has enacted. 11.2 per cent of the articles in the corpus are cited at least once in fifty thousand judgments; 187 instruments are never cited at all. The obvious objection — that the published record is commercial and the denominator is not — is answered in §5.5 by rebuilding the denominator from the jurisdiction clause of the Commercial Courts Law itself, under two readings, and by a fourth denominator that concedes the objection entirely. The share moves; the conclusion does not.

The sharpest form of the result needs no denominator at all, because it compares two instruments to themselves. The Commercial Courts Law, which tells the court how to proceed, has 97.9 per cent of its articles cited. The Civil Transactions Law, the codification of Saudi private law and the instrument the jurisdiction clause names by paragraph, has 25.8 per cent of its 721. The procedural instrument is nearly exhausted; the substantive code is barely opened. Both are within the same court's jurisdiction, both were enacted by the same legislature, and both are cited by the same judges in the same judgments.

Three further results follow. What is applied is overwhelmingly procedural (88.7 per cent of matched citations), and the most-cited provision in the entire corpus, cited 14,073 times, is the article stating which disputes the court may hear at all. The narrowing is not imposed by the parties: segmenting each judgment at its own headings shows the court’s reasons to be *more* procedural than the statement of the case, not less, and drawn from fewer instruments. And amendment activity predicts neither citation nor its absence, once the composition of the statute book is controlled for.

Three methodological points are reported here rather than left implicit, because each was a defect that produced confident wrong answers before it was found. Matching a citation to an instrument is not string containment: a regulation’s title contains its parent law’s title, and a matcher that accepts containment in either direction silently misdirected roughly half of all citations in an earlier version of this analysis (§4.2). Anaphora — ‘the Law’, ‘the Regulation’, ‘that same Law’ — must be resolved by kind rather than by recency. And the segment of a judgment before the reasons is not the parties’ pleadings: it is written by the court, and at least 44.3 per cent of the statutory citations in it sit in a sentence whose actor is the court (§4.4). Each of these would change a reported result, and none is visible in the output of a matcher that reports only a match rate.

The article proceeds as follows. §2 places the work in three literatures. §3 describes both corpora and what the publishing interface does and does not provide. §4 sets out extraction, matching, article-number parsing and segmentation. §5 reports the results and the denominator robustness. §6 states what the data cannot support, and §7 asks what a legal system that uses a ninth of its own statute book is telling us about codification.

2 Related work

Three literatures meet here, and the gap this article occupies is between them rather than inside any one of them.

Measuring legislation. A quantitative literature treats a statute book as an object with size, structure and change over time. Katz and Bommarito (2014) measure the complexity of the United States Code through its structure, language and interdependence; Coupette et al. (2021) follow statutes and regulations in the United States and Germany over time as evolving networks. This work is careful about the enacted text and, necessarily, silent about use: nothing in a statutory corpus records whether anyone ever invoked a provision. The machine-readable legal corpora that have grown alongside it — Henderson et al. (2022), Niklaus et al. (2024) — are assembled to train language models and hold running text rather than the article-level status and identity a join of this kind requires.

Measuring citation in judgments. A second literature treats citation as evidence of what courts actually rely on. Fowler et al. (2007) build the citation network of United States Supreme Court precedent and use its structure to measure the legal importance of individual cases. The unit there is the precedent, and the network is a court citing itself. Statutory citation by courts has attracted much less attention, in part because it is less structurally interesting — a court citing a statute makes an edge to a different kind of object — and in part because in common-law systems the interesting citation is to cases. In a recently codified civil-law jurisdiction the reverse holds: what a court cites is overwhelmingly statute, and the question of which statute is not decorative.

Selection: what reaches a court, and what is published. Any claim made from decided cases has to confront two filters. Priest and Klein (1984) model the first: disputes that settle differ systematically from those tried, so litigated cases are not a sample of disputes. Siegelman and Donohue (1990) document the second by comparing published against unpublished employment discrimination cases and finding them different in kind, not merely in number. Both filters apply here with full force, and neither can be removed by better extraction. They bound what the results mean rather than what they measure: this article measures what the published record contains, and §6 states plainly what that record is not.

The Saudi legal system. Scholarship on Saudi law has been overwhelmingly doctrinal and institutional, and for good reason: until recently there was nothing else to work with. Vogel (2000) remains the standard English-language treatment of the courts and their relation to Islamic legal doctrine, written when judgments were not published at all. The publication of commercial judgments at scale, and the codification programme of the last decade, have created materials that did not exist when that literature was written. This article is an attempt to use them quantitatively without pretending that the doctrinal questions have been answered by counting.

3 Data

3.1 The legislative corpus

The legislative side is a verified corpus of Saudi primary legislation: 15,855 articles distributed over 291 instruments, each article carried as a record with its number, its heading, its text, and its legal status — original, amended, added or repealed. Instruments are laws (*nizam*) and their implementing regulations (*la’iha tanfidhiyya*), which in Saudi practice are separate instruments with independently numbered articles, a distinction that turns out to matter for matching (§4.2). Provenance for each record — the route by which it was retrieved, the corroboration behind it, and any discrepancy between sources — is carried in the corpus itself rather than asserted in prose.

3.2 The judgment corpus

The judicial side is 50,666 judgments in full text, published by the Ministry of Justice and retrieved through the interface its own portal uses. Each record carries the judgment number, the court, the city, the Hijri date, and the text of the judgment as published, with the parties’ names already masked by the publisher.

Two features of the corpus bound everything that follows. It is overwhelmingly commercial: the commercial courts supply the great majority of the published judgments, first-instance and appellate, and other jurisdictions appear only in small numbers. And the interface exposes fields that would have made the analysis in §4.4 unnecessary — `judgmentFacts`, `judgmentReasons`, `judgmentRuling` — which are null in every one of the 50,666 records. The structure the courts write into their judgments is present; the structure the publisher promises is not.

4 Method

4.1 Extracting a citation

A Saudi judgment cites a provision in a stable form: the article, then the preposition *min* (‘of’), then the name of the instrument. Citations are extracted by matching that form and capturing two groups, the article expression and the instrument name. 106,337 citations are extracted from 39,336 of the 50,666 judgments (77.6 per cent); the remainder cite no statute in this form at all.

The capture needs a stopping rule. Judgments routinely quote the article they have just cited, and a name capture that runs to the end of the sentence swallows the quotation and matches nothing. Captures are therefore truncated at a list of tokens that begin a new syntactic unit — *al-sadir* (‘issued by’), *wa-llati nassat* (‘which provided’), *ala al-nahw al-ati* (‘as follows’) and others — before matching is attempted.

4.2 Matching an instrument

Instrument names in judgments differ from registry titles in predictable ways: the singular *nizam al-mahkama al-tijariyya* where the registry writes the plural *nizam al-mahakim al-tijariyya*; a regulation called *la’ihat nizam X* where the registry writes *al-la’iha al-tanfidhiyya li-nizam X*; and parenthetical

variants in the registry itself. Each registry title is expanded into the forms a judgment might use, and a captured name is matched against that index.

The direction of containment carries meaning, and getting it wrong produces confident errors rather than misses. A regulation's title contains its parent law's title as a substring, so a test that accepts containment in either direction will send a citation of article 16 of the Commercial Courts Law — the article stating the court's jurisdiction — to the Implementing Regulation of that same Law. In an earlier version of this analysis that single defect misdirected roughly half of all citations, and it was silent, because every citation still matched something. Matching therefore proceeds in three ordered steps: exact equality; then the longest registry title contained in the captured name, so that a citation spelling out the regulation's title reaches the regulation; then the shortest registry title containing the captured name, so that a citation naming only the law reaches the law.

Anaphora is resolved by kind. A bare *al-nizam* ('the Law') or *al-la'iha* ('the Regulation'), and phrases such as *dhat al-nizam* ('that same Law') or *al-la'iha nafsuha* ('the Regulation itself'), refer to an instrument named earlier in the same judgment. Resolving them to whichever instrument was named most recently sends articles of a law into its regulation whenever the regulation was mentioned last, which in a commercial judgment it usually was. *Al-nizam* is therefore resolved to the last *law* named in that judgment and *al-la'iha* to the last *regulation*. Anaphoric matches are recorded separately throughout: 94,269 citations name their instrument and 7,370 are resolved, and a resolved reference is a weaker observation that a reader should be able to set aside. 4.4 per cent of citations match no instrument in the registry and are dropped.

4.3 Reading an article number

Article numbers in Saudi judgments are more often spelled out than written in digits: *al-madda al-khamisa wa-al-tisun bad al-mi'a* is article 195. A parser that reads digits alone would measure a minority of citations and report it as a census. Article expressions are therefore parsed by a reader for Arabic ordinals that also handles Eastern Arabic numerals, compound forms, and both paragraph constructions — the compressed 16/3 and the expanded 'paragraph 3 of article 16' — returning an article number and, where present, a paragraph. 2.3 per cent of expressions cannot be parsed and are dropped. A further 1.5 per cent parse to a number beyond the last article of the matched instrument — a mismatch between the judgment and the registry, or a matching error — and are also dropped rather than assigned.

4.4 Separating the voices

A judgment is not one voice. It opens with a statement of the case, passes to the court's reasons, and closes with the operative disposition, and a statute cited in the first is not a statute the court applied. With the API's own fields empty, the segmentation is read from the headings the courts write: *al-waqai* (the statement of the case), then *al-asbab* (the reasons), then *hakamat al-daira* ('the circuit has adjudged'). Where the headings are present and in order the judgment is segmented; 27,350 judgments (54.0 per cent) qualify, and the rest are excluded from the voice analysis rather than forced into a shape they do not have. Because anaphora resolution needs document order, the text is scanned once and each citation is assigned to a segment by its offset.

One caution about the first segment matters enough to state as method. It is tempting to call it the pleadings, and wrong: *al-waqai* is written by the court, and it carries the court's own procedural narration alongside the parties' arguments as the court reports them.¹ Counting it as argument overstates the bar's citation practice and understates the bench's. Each citation in the segment therefore carries an attribution read from the sentence around it: whether the nearest actor to the citation is the court or a party. Because Arabic places the subject after the verb as readily as before it, the sentence is searched in both directions and the nearest cue wins.

¹A typical instance, in translation: 'The circuit notes that it held this preparatory hearing pursuant to article 90 of the Implementing Regulation of the Commercial Courts Law' — the court describing a step it took, not an argument put to it.

Table 1: Statutory citations by segment, across 27,350 segmented judgments.

Segment	Citations	Instruments	Procedural
Statement of the case	17,209	79	83.8%
Reasons	42,584	67	94.1%

The attribution cue is lexical, and its accuracy is reported rather than assumed. On a random sample of forty citations classified as the court's, 37 were the court narrating, two were ambiguous, and one was a party's memorandum written in the first person. The complementary class is much weaker, so no party share is reported anywhere in this article: what is reported is the court-attributed count, as a floor.

5 Results

5.1 How much of the statute book is applied

Of the 15,855 articles in the corpus, 1,774 are cited at least once in 50,666 judgments — 11.2 per cent of the statute book. At the level of the instrument, 104 of 291 instruments are cited at least once (35.7 per cent) and 187 are never cited at all.

The published record is heavily commercial, so a reader should expect criminal, family and administrative instruments to be absent. They are, and so is much else: entire commercial and financial instruments, with articles addressed to disputes that plainly arise, appear nowhere in fifty thousand judgments. Absence here is not proof that an instrument is unused; it is proof that its use, if any, is not visible in what the state publishes.

5.2 How concentrated the applied part is

The applied part is not merely small but concentrated inside itself. The single most-cited instrument accounts for 38.9 per cent of all matched citations, and the ten most-cited for 96.8 per cent. At article level the skew is sharper still: one article carries 14.4 per cent of all citations whose article number could be read, the top ten carry 43.1 per cent, and the top hundred 79.1 per cent.

What is applied is also overwhelmingly procedural. Instruments governing how a case is brought, heard, proved and enforced account for 88.7 per cent of matched citations; everything the law says about the substance of commercial obligations accounts for the rest.

The most-cited provision in the corpus, cited 14,073 times, is the article that lists the disputes the commercial court may hear at all. The provisions immediately behind it govern hearing a case in the absence of a defendant, the preparatory hearing, ordinary written evidence, and the plea of want of jurisdiction. The busiest provision in Saudi commercial adjudication is not a rule about commerce. It is the court's statement that it may proceed.

5.3 Who narrows: the bar or the bench

The obvious explanation for that narrowing is that courts answer what is put to them, and what is put to them is procedural. The segmentation tests it, and does not support it.

Table 1 shows the court's reasons to be more procedural than the statement of the case, not less, and drawn from a narrower set of instruments: 67 against 79. In 25 instruments a statute is cited in the statement of the case and never once in a court's reasons.

The attribution measurement sharpens this rather than softening it. Of the 17,209 citations in the statement of the case, at least 7,626 (44.3 per cent) sit in a sentence whose actor is the court itself. Those citations are 98.6 per cent procedural — the court reciting the provisions under which it convened, proceeded and satisfied itself of jurisdiction. The remainder, which is where the parties' arguments live, is 72.0 per cent procedural. Leaving the court's own narration inside the segment was flattering its procedural

Table 2: Article-level coverage under four denominators. The article-level join covers the instruments whose article records are on disk, so four instruments cited at instrument level but with no article number that parses into range do not appear here.

Denominator	Instruments	Articles	Ever cited
The whole registry	290	15,855	11.2%
Article 16 scope, broad	115	8,356	19.9%
Article 16 scope, narrow	54	5,297	29.4%
Instruments ever cited	100	8,954	19.8%

share; taking it out widens the gap between what is put to the court and what the court answers with, from 83.8 versus 94.1 to 72.0 versus 94.1.

The narrowing therefore happens at the bench, not at the bar. Substantive instruments are raised and are answered with procedure.

5.4 Amendment and application

If legislative attention followed judicial use, the amended part of the statute book would be the applied part. It is not.

At instrument level, over the 266 instruments whose records carry a status breakdown — of which only 88 are cited at all — the rank correlation between an instrument’s churn and its citation count is $\rho = .187$: weakly positive and carried by a handful of large instruments.

At article level the pooled comparison appears to say the opposite of a null. Across 6,767 articles with a recorded status, 25.5 per cent of the 518 amended, added or repealed articles are ever cited, against 21.2 per cent of the 6,249 original ones, and an amended article draws 40.1 citations on average against 11.4 — a ratio of $3.5\times$.

That ratio is a composition effect and should not be reported without its control. Amended articles are not spread evenly across the statute book, and the instrument carrying the largest share of all citations is also among the most heavily amended; the pooled figure mostly records which instruments are litigated, not which articles are amended. Controlling within the instrument — one vote per instrument, over the 44 instruments holding both amended and original articles — amended articles are more likely to have been cited in 19 instruments, less likely in 24, with 1 tie: a two-sided sign test gives $p = .542$. Inside an instrument, whether an article has been amended tells you nothing about whether a court has ever applied it.

5.5 Choosing the denominator

A share is only as good as what it is divided by, and the objection to the first result is obvious: the published judgments are overwhelmingly commercial, so a denominator holding the whole registry counts instruments a commercial court could never reach. Dividing by them makes the applied law small by construction.

The objection is answered by measurement, and the scope is drawn by the statute rather than by the author. Article 16 of the Commercial Courts Law enumerates what the court may hear, and its paragraphs name their instruments: partnership contracts under the Civil Transactions Law (§3), the Companies Law (§4), the Bankruptcy Law (§5), the intellectual property laws (§6), and ‘the other commercial laws’ (§7). Only the last requires judgement, so it is drawn twice — narrowly, as the instruments that constitute trade itself, and broadly, as everything whose subject is trade, finance or the regulation of commercial activity — and both readings are reported. Table 2 adds a fourth denominator that concedes the objection entirely: the instruments courts are observed to cite at all, which asks how much of an instrument is used even where the court does go.

The finding survives every denominator that can honestly be constructed. On the narrowest reading of the court’s own jurisdiction — 5,297 articles across 54 instruments, every one of them within Article 16 — 1,559 articles are ever cited: 29.4 per cent. Restricting to the instruments courts actually do cite leaves 19.8 per cent. The share moves with the denominator, as it must; what does not move is that most of the law the court is given goes unused.

Inside instruments the same gradient appears again, and it is sharper than anything at the level of the instrument. The Commercial Courts Law has 94 of its 96 articles cited, 97.9 per cent; then 88.4 per cent of the Evidence Law; 70.1 per cent of the Companies Law; 51.1 per cent of the Bankruptcy Law. And 25.8 per cent of the Civil Transactions Law — 186 of 721 articles — the codification of Saudi private law, and the instrument ¶3 of Article 16 points to by name.

That contrast is the result stated as sharply as the data allow. The instrument that tells the court how to proceed is almost exhausted; the instrument that tells it what the parties owe each other is barely opened.

6 Threats to validity

7 Discussion

8 Conclusion

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Availability of data and code

The Journal of Legal Analysis applies the data and code availability policy of the American Economic Association, and this article is written to it. The legislative corpus, the judgment corpus, the extraction and matching code, the ordinal parser, the segmentation and attribution code, and the scripts that produce every number in this article are public.² The judgments are reproduced as the Ministry publishes them,

²Repository: <https://github.com/Abdullah-m7/saudi-legal-corpus-ai>. Every measurement reported here is generated into a single file of macros by `make_numbers.py` and typeset from it; no number in this manuscript is typed by hand.

with parties' names masked at source; a further redaction pass over identifiers that the publisher's own masking missed is included in the repository and documented there.

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